

Sentiment Analysis and Text Classification in College Exit Interview Form

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Abstract—As students complete their journey at Jose Rizal University, they are required to provide feedback on the university's overall performance, not just academics. Graduating students and transferees must fill out a College Exit Interview Form to obtain essential documents like a good moral certificate. To streamline this process and assist the guidance office in analyzing and categorizing these forms efficiently, we are developing an AI model that will help improve productivity in assessing these documents.

C.E.I.F, Sentiment Analysis, Text Classification, Hugging face

I. INTRODUCTION

Natural Language Processing (NLP) includes the subfield of sentiment analysis, and NLP can be used to categorize and extract information with the aid of machine learning methods. Finding out if students are content with the amenities or services provided is important since they are consumers in an educational setting[1]. Analyzing course evaluations plays a crucial role in deciphering students' learning outcomes. By extracting sentiments from these evaluations, educators can gain a deeper understanding of students' experiences, identifying areas of improvement and tailoring teaching strategies to better meet students' needs [2]. Nonetheless, most institutions deal successfully with quantitative feedback, while qualitative feedback is either manually interpreted or ignored altogether [3]. To address this problem, we propose to analyze feedback automatically using sentiment analysis. Sentiment analysis is domain-dependent and although it has been applied to the educational domain before, it has not been previously used for real-time feedback [4]. Sentiments and opinions expressed by students are a valuable source of information not only for analyzing students' behavior towards a course, topic, or teachers but also for reforming policies and institutions for their improvement [5].

II. RELATED LITERATURE

Today sentiment analysis and text classification have made their way to analyzing the feedback and opinion by means of NLP. Since institutions strive to better the gist of their services, the assessment of the students' feedback particularly in the exit interview form has much significance in increasing of the qualities of education and institutional services. In this research, the current literature is reviewed in concern to sentiment analysis and text classification

techniques used in learning environments particularly in capturing student feedback.

A. Sentiment Analysis in Educational Feedback

In their work, Khaizer et al. (2023) applied text based classification and NLP on the students' feedback on Institutional facilities. The technique they used include machine learning and Natural Language Processing in which they sort sentiments into positive, negative and even neutral. It marks that, using sentiment analysis, one can obtain meaningful information about students in order to enhance institutional facilities. They show that using NLP and text classification techniques can help to process the amount of feedback very effectively and quickly, which is especially important in exit interview forms where a variety of opinions should be identified very fast[1].

Mahmood et al. (2022) used a lexicon-based technique for sentiment analysis on student feedback, focusing on how simple as lexicon based models may be, they are capable of producing high levels of sentiment classification. This proves that for them the lexicon formation relevant to student feedbacks can help them analyze the sentiment with high precision. When labeled training data is hard to come by, this method becomes doubly useful since lexicon based systems are dependent on a list of sentiment terms. This technique can be used in college exit interviews where qualitative feedback means that efficient sentiment analysis is needed for better decisions[9].

B. Machine Learning Techniques in Text Classification

Sidorov et al. (2014) discussed the application of machine learning approach in text classification, and this particular research opt for short text classification. Their study may be most suited to college exit interview forms where responses are short. This means that Naïve Bayes and Support Vector Machines (SVM) enhanced the classification rate according to the research results. Combined with feature extraction methodologies such as Term Frequency-Inverse Document Frequency (TF-IDF), the application of the employed algorithms ensured fine-grained classification, even in environments where a small amount of textual data was produced[10].

In another older study, Kumari et al. (2022) introduced deep learning based hybrid model for text analysis for classification of the student feedback. CNN was combined with LSTM and the result was quite successful for classification in text data. Their study also points toward the applicability of deep learning models in handling sentiment analysis in a bid to identify more complex sentiments from students' feedback. Since the responses elicited during exit interviews are rich and diverse, hybrid models will enhance the classification performance since they allow for semantic as well as contextual consideration of word use[11].

C. Application in Large Scale Feedback Systems in Schools

Scalable sentiment analysis systems of the educational institutions were discussed by Mehta et al., (2020). They came up with a model that can cater to big data to ensure that the rate of feedback processing and analysis in real-time. The research can be valuable to educational organizations and institutions trying to make changes in response to substantial student feedback data, for example, an exit interview. This research underscores the importance of scalability and real-time processing when putting in place sentiment analysis systems in education[12].

Finally, in their study, Garcia et al., 2023 examined the opportunities and issues of big data for sentiment analysis of educational organizations. Their research was to evaluate how big data analytics can support huge collections of unstructured data like student feedback, and how sentiment analysis can be included with advanced analytics. This work implies that the implementation of big data solutions along with implementing sentiment analysis methods in educational settings can enhance the analysis and penetrate deeper into the institution to foster the right decision-making[13].

III. METHODOLOGY

This section discusses the various details of the study's implementation. It includes the proposed model, the data sources, and the timeline.

A. Model

The researchers propose using Bidirectional Encoder Representations from Transformers (BERT) as the Text Classification and Sentiment Analysis model. The said model is proven effective when it comes to text classification and sentiment analysis [6][7].

The BERT model that the researchers will utilize is from Google's BERT available from the HuggingFace library [8].

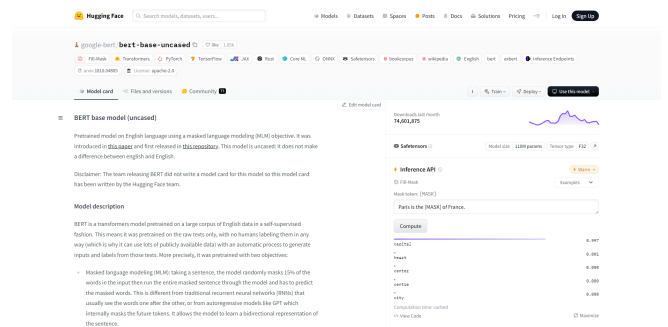


Figure 1. google-bert/bert-base-uncased model

B. Data Sources

• Training Data

The researchers will use the "Sp1786/multiclass-sentiment-analysis-dataset" from HuggingFace to train the model for sentiment analysis. It is a multi-class sentiment analysis that has **41,643 rows**, containing data for text, label, and sentiment.

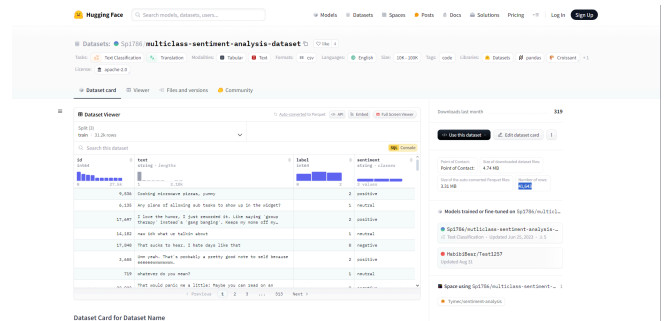


Figure 2. Sp1786/multiclass-sentiment-analysis-dataset

The researchers will fine-tune the model by using the College Exit Interview form data. It includes the sentiments of each transferee and graduating student about the overall college experience and their reason for leaving if they are transferring.

• Testing Data

The case in our testing data will be the same with our training data since the sentiment pre-trained model does offer vast amount of data for testing.

• Validation Data

The said data for the validation we will be using the data that we will be requesting from the guidance and will be used as well to predict whether the model we generate will be accurate.

C. Timeline

The Gantt chart will illustrate a week-by-week timeline for each process, including Data Collection, Data Preparation, Data Cleaning, and others. We can refer to it in the figure below.

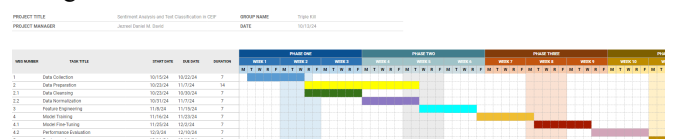


Figure 3. Gantt Chart Timeline

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